IT227 – Computer Systems Programming

Course Placement: Computer Systems Programming is a core course for second-year students of the BTech ICT program.

Course format: It is **3 hours lecture** and **2 hours lab practical** every week.

Prerequisite: IT112 and IT113 Introduction to Programming (C Programming), IT121 Digital Logic Design and Computer Organization.

Course Content: Course objective: The course takes an introductory look at the core abstractions in operating systems: processes, virtual memory and files. It takes an in-depth look at the OS services provided by system calls, how system calls work, and how they can be used. Students will become familiar with writing application programs using system calls.

Text Books:

- Bryant and O'Hallaron, Computer Systems: A Programmer's Perspective, 3rd edition, Pearson India, 2016.
- Operating Systems: Three Easy Pieces, Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau, 2018, Version 1.00. (Freely available online.)

Assessment method:

- Lab Programming Assignments: 20%
- 1st In-Semester Exam: 15%
- 2nd In-Semester Exam: 20%
- End-Semester Exam: 25%
- Project: 20%

Course Outcomes:

After successful completion of the course, students will have the ability to

- Understand the concept of virtualization, operating systems abstraction concepts such as process and thread, and more advance concepts such as interprocess communication using signals, race codition, mutual exclusion problem and solution using various locking mechanisms.
- Acquire practical knowledge by developing parallel applications using multiprocess and multithreading, performing interprocess communication in systems with shared-memory architecture using signals.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
	X	X									X

Lecture Schedule:

SI. No.	Description	No. of Lectures
1	Introduction: System Programming Introduction	3

	Key OS abstractions: processes, virtual memory and files, virtual address space, system calls, interrupts, user and kernel mode, process state transition, context switching, saving and restoring context	
2	Process creation, process termination, reaping child processes, putting processes to sleep, loading and running programs, Unix shell, IPC Pipes	5
3	Signal terminology, sending signals, receiving signals, normal and abnormal termination, signal blocking, job control using signals	5
4	Address translation, segmentation, page tables, TLB, page fault control flow, page replacement policies, Belady's anomaly, thrashing, case study: Linux VM system	5
5	Opening and closing files, Unbuffered I/O vs buffered I/O, directories, file metadata, file sharing, symbolic link, I/O redirection	5
6	Thread creation, thread termination, reaping terminated threads, thread memory model, shared variables, race conditions	5
7	Mutual exclusion problem, solutions to mutual exclusion problem using locks/semaphores, deadlocks, necessary conditions for deadlock, dining-philosophers problem, producerconsumer problem, readers-writers problem	5
8	Network Programming: Communication Layers (Network, Transport), Protocols basics: Internet Protocol (IP), TCP: connection-oriented, UDP: connectionless, standard services and assigned ports, client-server communication using sockets	5